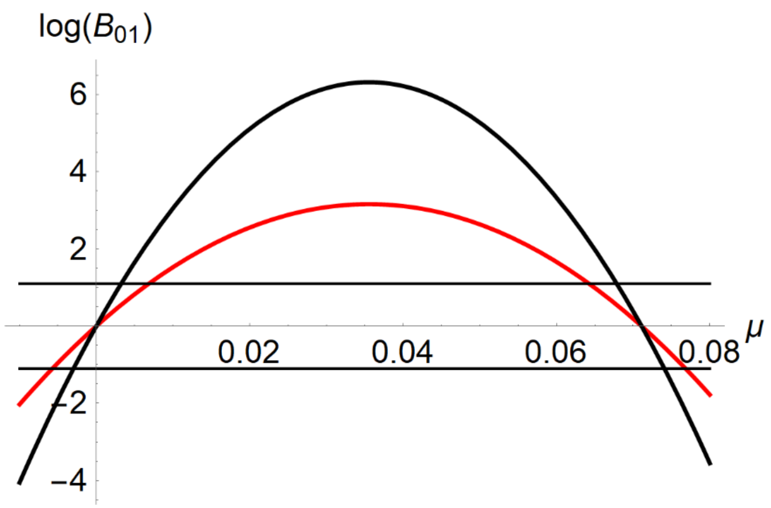

Figure 2. Log of Bayes factor as a function of for n = 5,000 and 10,000

Given n and q, if $R(\bar{X},n)=q$ for some $\bar{X}$ then for $\bar{X}$ closer to 0 $R(\bar{X},n)>q$. However, $B(0,\mu,\bar{X},n) > q$ holds only for values of μ in the neighbourhood around $\bar{X}$ as explained earlier. For every $\bar{X}$ in the appropriate range, there is a minimum n required to satisfy $B(0,\mu,\bar{X},n) > q$. So a reasonable question is how the minimum n covaries with $\bar{X}$ throughout this range. For any benchmark sample size, n_b, we can establish the subrange of $\bar{X}$ in which the minimum required n is exceeded by the benchmark n_b. That will then give us the range of $R(\bar{X},n)$ for which, given n and q, it also is the case that $B(0,\mu,\bar{X},n) > q$ for at least some values of μ around $\bar{X}$.

In the example, n = 5000 and q = 3, so inverting equation (5) gives $\bar{X}=\sqrt{2\log(q)/n}=0.0209629$. At that value of $\bar{X}$, $R(\bar{X},n)=23.5778$, so this is the upper bound on B_{R01} when n = 5000 and q = 3 for which there is a neighbourhood of μ in which $B(0,\mu,\bar{X},n) > 3$. The resulting subrange of $\bar{X}$ is [0.0209629, 0.035557], and these are the sample means for which $3 \le R(\bar{X},n) \le 23.5778$. Thus, for a sample of 5000 obtaining a

Robert BF anywhere in the range $3 \le R\left(\bar{X},n\right) \le 23.5778$ also implies that $B\left(0,\mu,\bar{X},n\right) > 3$ for at least some values of μ around $\bar{X}$, thereby constituting a Lindley case.

Note that we obtain similar subranges of $\bar{X}$ for the Rouder et al. t-tests, with the same lower limit as the one just derived. Recall that the upper limits of $\bar{X}$ for which the JSZ and scaled-information BFs exceed 3 are 0.03489 and 0.03356 respectively. At the lower limit of $\bar{X} = 0.0209629$, the JSZ BF = 20.9126 and scaled-information BF = 16.6752. Thus, for instance, given a sample of 5000 and JSZ BF between 3 and 20.9126, we know that it will be a Lindley case.

Our next concerns are threefold. First, how does this Lindley case region for $R\left(\bar{X},n\right)$ or the Rouder et al. t-test BFs behave as *n* increases? Second, for any particular *n*, how likely is it that even when H_0 is <u>true</u>, nonetheless Lindley cases will occur? Third, how does that probability covary with *n* or *q*?

Starting with the first concern, equation (11) tells us that as *n* increases the threshold $\bar{X}$ will decrease. Substituting the inverted version of equation (11) into equation (4) gives this version of $R\left(\bar{X},n\right)$:

$$R\left(\bar{X},n\right) = \sqrt{n+1}\exp\left(-\frac{n2\log(q)}{2n+2}\right) \tag{12}$$

Differentiating it with respect to *n* yields

$$\frac{\partial R\left(\bar{X},n\right)}{\partial n} = \frac{q^{\frac{1}{n+1}-1}(n-2\log(q)+1)}{2(n+1)^{3/2}} \tag{13}$$

which is positive for $n > 2\log\left(q\right)-1$. Therefore, although the threshold $\bar{X}$ declines with increasing *n*, the upper bound on $R\left(\bar{X},n\right)$ nevertheless increases.

To address the second concern, let us return to our example. We have $n = 5000$ and $q = 3$, and inverting equation (5) gives $\bar{X} = \sqrt{2\log\left(q\right)/n} = 0.0209629$. As observed earlier, the upper value of the appropriate range for $R(\bar{X}, n)$ is $\bar{X} = 0.035557$. Likewise there is a corresponding interval below 0, [-0.035557, -0.0209629]. The standard error of the mean is $1/\sqrt{n}$, so it turns out that the areas under the normal curve for these two intervals sum to approximately 0.1263. Thus, the probability of simultaneously confirming and disconfirming H_0 (i.e., observing a Lindley case) when H_0 is true is 0.1263. The same argument applies to the Rouder et al. t-tests. The areas of their intervals under the normal curve turn out to be 0.1246 for the JZS and 0.1206 for the scaled-information tests. These are not negligible probabilities.

Regarding the third concern, the probability of observing a Lindley case when H_0 is true increases with n. For $R(\bar{X}, n)$, at n = 10,000 the probability is 0.130 and at n = 20,000 the probability is 0.133, and it is easy to show that it asymptotes at about 0.138. In general, the asymptote is $2 - \mathrm{erfc}\left(-\sqrt{\log(q)}\right)$. On the other hand, we can see that it decreases as q increases (e.g., when q = 10 the asymptote is 0.032). Nevertheless, these findings suggest that for values of q regarded by many researchers as "substantial", the probability of Lindley cases is non-negligible even when H_0 is true, and moreover it increases with sample size.

Therefore, point nulls are not "provable" by Bayes factors. Instead, if we want to find the "best bet" on θ then examining $B(\theta, \theta_0, x)$ (or at least the full range of $ER(\theta$, x)) makes more sense than restricting attention to $ER(\theta_0$, x) or the corresponding BF.

Quandary at the Boundary

Unfortunately, it is not difficult to show that with large samples support-rejection regions can become problematic in an additional way to Lindley cases. Given a big enough sample, a value arbitrarily close to the point null may be "rejected" by support-interval logic while the point-null will be "supported" at the same criterial ratio. For instance, given a sample size of